SEVA INNOVATION CHALLENGE 2026 • MAHARASHTRA ZONE PITCH DECK

GrievanceIQ

Voice-first, multilingual autonomous public banking grievance redressal and predictive operational intelligence platform for Maharashtra.

Project Overview

- Institution: Sardar Patel Institute of Technology (SPIT), Mumbai
- Team: Vedant Patil, Ashmit Patil, Mayuri Tarde | Mentor: Dr. Rupali Sawant
- Target Impact: 65M+ Citizens, Farmers (PM-KISAN) & Women (Ladki Bahin)
- Live Working Prototype: https://grievanceiq-sigma.vercel.app
- System Maturity: Fully Functional Prototype | 100% DPDP Act Compliant

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The Problem

What real challenge in society, government service or industry are you solving?

What is the problem?

Digital banking disputes (UPI timeouts, ATM non-dispensations, DBT welfare delays) face 24 to 72 hours of manual triage delay in banks. Crucially, portals lack Marathi voice input, disenfranchising rural citizens who speak regional languages. Furthermore, plain-text retention of unmasked Aadhaar, Debit Card, and PAN numbers violates India's DPDP Act 2023.

Who is affected, and where?

Over 65 million retail depositors, farmers (PM-KISAN), women (Ladki Bahin DBT), and MSMEs across all 36 districts of Maharashtra (especially rural talukas and semi-urban centers), along with 3,500+ rural cooperative bank branches overwhelmed by manual grievance backlogs.

Evidence

RBI Integrated Ombudsman Report 2023 records 703,000+ complaints nationally (45%+ in digital/ATM banking). In Maharashtra alone, over 12 Crore active bank accounts generate over 1.2M transaction disputes annually, averaging 30 days for manual redressal.

Current gaps

Existing portals (CPGRAMS, RBI CMS, bank helpdesks) require complex English text entry, lack automated ML triage, expose raw citizen PII in logs/databases, and provide zero predictive queue forecasting for administrative supervisors.

65M+

Citizens, farmers & MSMEs affected across Maharashtra's 36 districts

Why it matters

Severe economic distress for rural families waiting weeks for stuck funds, loss of trust in digital DBT welfare, and heavy legal penalties under DPDP Act 2023.

3 Our Solution & Innovation

What have you built or designed, how does it work, and what is new about it?

Our solution in brief

GrievanceIQ is an autonomous public grievance platform featuring real-time continuous Marathi/Hindi voice dictation, ingestion-boundary DPDP PII scrubbing, 89.2% accurate ML triage (15k dataset), dynamic QR-coded statutory PDF receipts, and predictive M/M/c Erlang-C queue capacity modeling with XAI documentary evidence verification.

How it works

1 Input

Citizen dictates dispute in Marathi/Hindi or enters text + uploads receipt proof.

2 Process

API scrubs PII (DPDP 2023), MNB classifies category (<30ms), OCR validates receipt.

3 Output

Instant Complaint ID, scannable QR statutory PDF receipt, and predictive queue dispatch.

What makes it new?

	Existing approach	Our approach
Cost	High licensing / Manual staff overhead	Zero license (100% open-source stack)
Speed / time	24 - 72 hours manual triage delay	< 30 ms autonomous classification
Reach / access	English text entry only	100% Trilingual (Marathi/Hindi/EN) + Voice
Ease of use	Complex forms & physical bank queues	Zero-latency hands-free voice dictation

Key USP

The only voice-first vernacular redressal platform with built-in DPDP Act 2023 PII scrubbing, dynamic QR receipts, and predictive M/M/c Erlang-C queue forecasting.

4 Feasibility & Implementation

How ready is the solution, and what will it take to implement it on the ground?

Current stage (mark ✓ where you are today)

- Idea
- Proof of Concept
- Working Prototype [✓]
- Field Pilot
- Ready to Deploy [✓]

Technology

React 18, Vite, Tailwind CSS, W3C Web Speech API, Python FastAPI (ASGI), Scikit-Learn (MultinomialNB, TF-IDF), EasyOCR, ReportLab 4.0 (ISO 18004 QR), SQLite (WAL mode). Fully containerizable with Docker for cloud or on-premise servers.

Estimated cost

Zero software licensing cost (100% open-source stack). Operational processing cost < Rs 0.15 per grievance. Pilot deployment budget: Rs 1.5 - 2.5 Lakhs for state-level server hosting, bandwidth, and verification testing.

Timeline

- M1 - M3: Pilot deployment across 5 rural cooperative & lead district banks in Maharashtra.
- M4 - M6: Integration with state DBT subsidy portals (Ladki Bahin, PM-KISAN) & CBS mock adapters.
- M7 - M12: Full statewide rollout across 36 districts with 2-way conversational voice bot.

Partners

Dept. of Cooperation & Banking (Govt of Maharashtra), State Level Bankers' Committee (SLBC Maharashtra), Maharashtra State Co-operative Bank, Regional Rural Banks (RRBs), and CSC e-Governance Centers.

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Impact & Scalability

What change will your solution create, and how can it grow beyond the first location?

1.2M+

Beneficiaries reached

Citizens & farmers reached in Year 1 across Maharashtra

95%

Saving in cost / time

Triage time slashed from 48 hrs to < 1 sec; 40% faster resolution

3,500+

Jobs or livelihoods

Frontline banking & CSC operators empowered with AI triage

Social impact

Democratizes financial justice for rural farmers, women, and non-literate citizens through spoken Marathi voice. Eliminates lost daily wages from physical bank queues and prevents identity theft by 100% PII masking under DPDP Act 2023.

Sustainability

Sustained via institutional SaaS / departmental IT allocations (~Rs 2 - Rs 5 per resolved claim), CSR grants from public/private financial institutions, or state e-governance administrative budgets.

Scale-up plan

Microservices architecture enables seamless horizontal scaling across all 36 districts of Maharashtra, other Indian states, and adjacent public departments (e.g., municipal corporations, revenue, public distribution system) by adding language dictionaries.

6 Team/Individual & Support Required

Who is behind the idea, and what help do you need from the Challenge?

VP — Vedant Patil
Full-Stack Architect & AI Lead
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AP — Ashmit Patil
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Thank you | vedant.patil24@spit.ac.in • https://grievanceiq-sigma.vercel.app

Support we are seeking

- ☑ Mentorship & technical guidance
- ☑ Pilot with a government department (Maharashtra)
- ☑ Funding / grant support for cloud scaling
- ☑ Incubation & workspace
- ☑ Testing, certification & regulatory compliance

Challenge support will connect GrievanceIQ directly with Maharashtra's Dept of Cooperation, SLBC & rural cooperative banks for on-ground pilot deployment and citizen impact.